

WEEK 3: SELLER AND INVENTORY MANAGEMENT SYSTEM

Scenario

The e-commerce company sells products supplied by multiple sellers. The system maintains seller details, tracks the products supplied by each seller, and monitors inventory levels to ensure product availability.

1. Create Seller and Inventory Tables

The **Seller** table is created to store the details of sellers, including Seller ID, Seller Name, Email, Phone Number, and Address.

The **Inventory** table is created to maintain the inventory details of products, including Inventory ID, Product ID, Seller ID, Stock Quantity, Stock Status, and Last Updated date.

```
SQL> CREATE TABLE Seller(  
2     Seller_ID VARCHAR(7),  
3     Seller_Name VARCHAR(25),  
4     Email VARCHAR(30) NOT NULL,  
5     Phone NUMBER(10),  
6     Address VARCHAR(45),  
7     CONSTRAINT PK_Seller PRIMARY KEY(Seller_ID),  
8     CONSTRAINT Un_mail UNIQUE>Email)  
9 );
```

Table created.

```
SQL> CREATE TABLE Inventory(  
2     Inventory_ID VARCHAR(7) PRIMARY KEY,  
3     Product_ID VARCHAR(7),  
4     Seller_ID VARCHAR(7),  
5     Stock_Quantity NUMBER(11),  
6     Stock_Status VARCHAR(11),  
7     Last_Updated DATE,  
8     CONSTRAINT FK_Prod  
9         FOREIGN KEY(Product_ID)  
10        REFERENCES Product(Product_ID),  
11     CONSTRAINT FK_Seller  
12        FOREIGN KEY(Seller_ID)  
13        REFERENCES Seller(Seller_ID),  
14     CONSTRAINT CK_Quan  
15        CHECK(Stock_Quantity >= 0)  
16 );
```

Table created.

2. Establish Relationships

Seller and Product

One seller can supply multiple products, and a product can be supplied by one or multiple sellers.

Relationship:

Seller → Product is **One-to-Many**.

Product and Inventory

Each product has inventory information, and the inventory table is used to track the available stock of each product.

Relationship:

Product → Inventory is **One-to-One**.

3. Constraints

Primary keys are used to uniquely identify the records in the tables.

The **Seller ID** is used as the primary key for the Seller table, while the **Inventory ID** is used as the primary key for the Inventory table.

The Inventory table also contains foreign keys to establish relationships with the Product and Seller tables.

A unique constraint is applied to the **Email** field to ensure that each seller has a unique email ID.

A check constraint is applied to **Stock Quantity** to ensure that the quantity does not contain a negative value.

4. Insert Values into Seller Table

Five seller records are inserted into the Seller table. The records contain the seller ID, seller name, email, phone number, and address.

```
SQL> INSERT INTO Seller VALUES ('S001','Auto Parts India','autoparts@gmail.com','9876543210','Chennai');
1 row created.

SQL> INSERT INTO Seller VALUES ('S002','Motor World','motorworld@gmail.com','9876543211','Tambaram');
1 row created.

SQL> INSERT INTO Seller VALUES ('S003','Speed Auto Supplies','speedauto@gmail.com','9876543212','Guindy');
1 row created.

SQL> INSERT INTO Seller VALUES ('S004','Gear House','gearhouse@gmail.com','9876543213','Porur');
1 row created.

SQL> INSERT INTO Seller VALUES ('S005','Drive Accessories','driveacc@gmail.com','9876543214','Chromepet');
1 row created.
```

```
SQL> SELECT * FROM Seller;
```

SELLER_	SELLER_NAME	EMAIL	PHONE
ADDRESS			
S001	Auto Parts India Chennai	autoparts@gmail.com	9876543210
S002	Motor World Tambaram	motorworld@gmail.com	9876543211
S003	Speed Auto Supplies Guindy	speedauto@gmail.com	9876543212
ADDRESS			
S004	Gear House Porur	gearhouse@gmail.com	9876543213
S005	Drive Accessories Chromepet	driveacc@gmail.com	9876543214
S006	Metro Auto Parts	metroauto@gmail.com	9876543215

```
SQL> INSERT ALL
 2 INTO Inventory VALUES ('I001','P2001','S001',12,'Available',TO_DATE('25-09-2026','DD-MM-YYYY'))
 3 INTO Inventory VALUES ('I002','P2002','S002',20,'Available',TO_DATE('25-09-2026','DD-MM-YYYY'))
 4 INTO Inventory VALUES ('I003','P2003','S003',18,'Available',TO_DATE('25-09-2026','DD-MM-YYYY'))
 5 INTO Inventory VALUES ('I004','P2004','S004',10,'Available',TO_DATE('25-09-2026','DD-MM-YYYY'))
 6 INTO Inventory VALUES ('I005','P2005','S005',14,'Available',TO_DATE('25-09-2026','DD-MM-YYYY'))
 7 INTO Inventory VALUES ('I006','P2006','S006',7,'Available',TO_DATE('25-09-2026','DD-MM-YYYY'))
 8 INTO Inventory VALUES ('I007','P2007','S007',9,'Available',TO_DATE('25-09-2026','DD-MM-YYYY'))
 9 INTO Inventory VALUES ('I008','P2008','S008',11,'Available',TO_DATE('25-09-2026','DD-MM-YYYY'))
10 INTO Inventory VALUES ('I009','P2009','S009',15,'Available',TO_DATE('25-09-2026','DD-MM-YYYY'))
11 INTO Inventory VALUES ('I010','P2010','S010',0,'Out Stock',TO_DATE('25-09-2026','DD-MM-YYYY'))
12 SELECT * FROM dual;

10 rows created.
```

```
SQL> SELECT * FROM Inventory;
```

INVENTO	PRODUCT	SELLER_	STOCK_QUANTITY	STOCK_STATU	LAST_UPDA
I001	P2001	S001	12	Available	25-SEP-26
I002	P2002	S002	20	Available	25-SEP-26
I003	P2003	S003	18	Available	25-SEP-26
I004	P2004	S004	10	Available	25-SEP-26
I005	P2005	S005	14	Available	25-SEP-26
I006	P2006	S006	7	Available	25-SEP-26
I007	P2007	S007	9	Available	25-SEP-26
I008	P2008	S008	11	Available	25-SEP-26
I009	P2009	S009	15	Available	25-SEP-26
I010	P2010	S010	0	Out Stock	25-SEP-26

```
10 rows selected.
```

6. Report 1: Seller-wise Product Report

The Seller-wise Product Report displays the details of products supplied by each seller.

The report contains:

- Seller Name
- Number of Products Supplied
- Available Stock

The report uses the Seller and Inventory tables to obtain the required information.

From the result, each listed seller has supplied one product, while the available stock varies according to the inventory records.

```
SQL> SELECT s.Seller_Name, COUNT(i.Product_ID) AS Product_Count, SUM(i.Stock_Quantity) AS Available_Stock FROM Seller s JOIN Inventory i ON s.Seller_ID=i.Seller_ID GROUP BY s.Seller_Name;
```

SELLER_NAME	PRODUCT_COUNT	AVAILABLE_STOCK
Auto Parts India	1	12
Motor World	1	20
Speed Auto Supplies	1	18
Gear House	1	10
Drive Accessories	1	14
Metro Auto Parts	1	7
Car Care Hub	1	9
Road King Supplies	1	11
Prime Motors	1	15
Auto Zone	1	0

10 rows selected.

7. Report 2: Stock Availability Report

The Stock Availability Report provides information about the current stock of products.

It displays:

- Product Name
- Stock Quantity
- Stock Status

The report helps to identify whether the products are currently available or unavailable and shows their corresponding stock quantities.

```
SQL> SELECT p.Product_Name, i.Stock_Quantity, i.Stock_Status FROM Product p JOIN Inventory i ON p.Product_ID=i.Product_ID;
```

```
PRODUCT_NAME
```

```
-----  
STOCK_QUANTITY STOCK_STATU
```

```
-----  
Brake Pad Set  
12 Available
```

```
Castrol Engine Oil  
20 Available
```

```
Car Phone Holder  
18 Available
```

```
PRODUCT_NAME
```

```
-----  
STOCK_QUANTITY STOCK_STATU
```

```
-----  
Bike Crash Guard  
10 Available
```

```
Steel Helmet  
14 Available
```

```
Exide Car Battery  
7 Available
```

```
PRODUCT_NAME
```

```
-----  
STOCK_QUANTITY STOCK_STATU
```

```
-----
```

```
Leather Seat Cover  
9 Available
```

```
Premium Floor Mat  
11 Available
```

```
LED Headlight  
15 Available
```

```
PRODUCT_NAME
```

```
-----  
STOCK_QUANTITY STOCK_STATU
```

```
-----  
New Car Perfume  
0 Out Stock
```

```
10 rows selected.
```

8. Report 3: Inventory Analysis

The Inventory Analysis report is used to analyse the overall stock maintained in the inventory.

The analysis includes:

Total Products Available

The total number of inventory records with **Available** status is **4**.

Products Out of Stock

The number of products with **Out of Stock** status is **0**.

Highest Stocked Product

The product with the highest stock quantity is **Running Shoes**, with a stock quantity of **5**.

Average Inventory Quantity

The average inventory quantity is **3.8**.

```
SQL> SELECT COUNT(*) AS Total_Products_Available FROM Inventory WHERE Stock_Quantity > 0;

TOTAL_PRODUCTS_AVAILABLE
-----
9

SQL> SELECT COUNT(*) AS Out_of_Stock_Products FROM Inventory WHERE Stock_Quantity = 0;

OUT_OF_STOCK_PRODUCTS
-----
1

SQL> SELECT p.Product_Name, i.Stock_Quantity FROM Product p JOIN Inventory i ON p.Product_ID=i.Product_ID WHERE i.Stock_Quantity=(SELECT MAX(Stock_Quantity) FROM Inventory);

PRODUCT_NAME
-----
STOCK_QUANTITY
-----
Castrol Engine Oil
20

SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory_Quantity FROM Inventory;

AVERAGE_INVENTORY_QUANTITY
-----
11.6
```

Conclusion

The Seller and Inventory Management System successfully maintains seller details and inventory information. The reports help to identify seller-wise products, monitor stock availability, and analyse inventory quantities. This system provides useful information for effective inventory management.